

SMS Spam & Phishing Detection

Using Machine Learning

Technical Report

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Executive Summary

This report presents the design, implementation, and evaluation of a machine learning-based SMS spam and phishing detection system developed by Team CP2077. The proliferation of unsolicited and malicious SMS messages poses a significant threat to mobile users worldwide, with phishing attacks increasingly exploiting SMS as a delivery vector for credential theft and financial fraud.

Our system leverages the well-established SMS Spam Collection Dataset, comprising 5,574 messages labeled as either legitimate (ham) or spam. We applied a comprehensive natural language processing pipeline including text cleaning, TF-IDF vectorization, and domain-specific feature engineering to prepare the data for classification. Five machine learning algorithms were trained and evaluated: Naive Bayes, Logistic Regression, Random Forest, Gradient Boosting, and Linear SVM.

The Random Forest classifier emerged as the top-performing model, achieving an accuracy of 98.39%, a precision of 99.25%, a recall of 88.59%, and an F1-score of 93.62%. These results demonstrate that ensemble methods combined with effective text preprocessing can reliably detect spam and phishing messages with minimal false positives. The system has direct real-world applicability in mobile network filtering, enterprise communication security, and consumer spam protection applications.

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1. Introduction & Problem Statement

The rapid expansion of mobile communication has made SMS one of the most widely used channels for both personal and commercial messaging. However, this ubiquity has also attracted malicious actors who exploit SMS to deliver spam, phishing attacks, and social engineering schemes. According to industry reports, billions of spam SMS messages are sent globally each year, resulting in financial losses, identity theft, and erosion of user trust in mobile platforms.

Spam SMS messages range from unsolicited advertisements to sophisticated phishing attempts that impersonate banks, government agencies, and popular services. Smishing — phishing via SMS — has grown substantially as users increasingly conduct financial transactions on mobile devices. Traditional rule-based filtering approaches are insufficient against evolving attack patterns, motivating the need for adaptive machine learning solutions.

1.1 Problem Statement

The core problem addressed in this project is the binary classification of SMS messages as either spam or legitimate (ham). Given a raw SMS message, the system must accurately determine whether it constitutes spam or phishing content, while minimizing both false positives (legitimate messages flagged as spam) and false negatives (spam messages that evade detection).

1.2 Objectives

- Design and implement an end-to-end SMS spam detection pipeline using machine learning.
- Compare the performance of five classification algorithms on the SMS Spam Collection Dataset.

- Engineer domain-specific features to improve detection accuracy.
- Evaluate models using appropriate metrics for imbalanced classification tasks.
- Identify the best-performing model suitable for real-world deployment.

1.3 Real-World Applications

The developed system has practical applications across multiple domains including mobile network operator spam filtering, enterprise SMS gateway security, consumer mobile security applications, and fraud detection systems in banking and e-commerce platforms.

2. Literature Review & Related Work

The problem of SMS spam detection has been studied extensively in the machine learning and natural language processing literature. We review the most relevant prior work to contextualize our approach and identify opportunities for improvement.

2.1 Early Approaches

Almeida et al. (2011) introduced the SMS Spam Collection Dataset used in this study and demonstrated baseline classification results using Naive Bayes, achieving accuracy rates around 97% with simple bag-of-words features. Their work established the benchmark for subsequent research and highlighted the class imbalance challenge inherent in spam datasets, where legitimate messages significantly outnumber spam.

Hidalgo et al. (2012) extended this work by exploring a range of content-based features including token frequencies, character n-grams, and structural message properties. They found that combining lexical and structural features consistently outperformed single-feature approaches across multiple classifiers.

2.2 Machine Learning Methods

Sahami et al. (1998) pioneered the application of Naive Bayes to email spam filtering, establishing the theoretical foundation for probabilistic text classification in spam detection. While originally designed for email, their framework translated effectively to SMS due to the similar bag-of-words assumption underlying both domains.

Drucker et al. (1999) demonstrated the effectiveness of Support Vector Machines (SVM) for text classification tasks including spam filtering. SVMs were shown to handle high-dimensional TF-IDF feature spaces effectively, achieving strong performance even with limited training data.

Breiman (2001) introduced the Random Forest algorithm, which has since become one of the most widely applied ensemble methods in spam detection. Its ability to reduce overfitting through bagging and random feature selection makes it particularly well-suited for text classification with noisy features.

2.3 Deep Learning Approaches

Rao & Gupta (2021) applied Long Short-Term Memory (LSTM) networks to SMS spam detection, demonstrating that sequential models can capture contextual relationships between words that bag-of-words models miss. Their approach achieved F1-scores above 98% but required significantly more computational resources than traditional ML methods.

Vinayakumar et al. (2018) explored convolutional neural networks (CNNs) for spam detection across multiple communication channels including SMS and email. They found that CNNs with word embeddings outperformed traditional ML methods on large datasets but offered marginal gains on smaller corpora such as the SMS Spam Collection.

2.4 Feature Engineering

Gomez Hidalgo et al. (2006) conducted a comprehensive study of feature types for spam filtering, including token presence, token frequency, character n-grams, and structural features such as message length and punctuation density. Their findings support the inclusion of hand-crafted features alongside TF-IDF representations, which motivated our feature engineering approach.

Delany et al. (2012) specifically addressed the SMS spam detection problem and highlighted the importance of domain knowledge in feature construction, noting that spam messages frequently contain URLs, excessive punctuation, and urgency cues that can be captured as binary or count features.

2.5 Handling Class Imbalance

Chawla et al. (2002) introduced the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic samples of the minority class to address dataset imbalance. In the context of SMS spam detection, where spam typically constitutes only 13-15% of messages, techniques like SMOTE and class-weighted loss functions have been shown to improve recall on the minority spam class.

2.6 Comparison Table

Author(s)	Year	Method	Best Accuracy	Key Contribution
Almeida et al.	2011	Naive Bayes	97.0%	Introduced SMS Spam Collection Dataset
Hidalgo et al.	2012	SVM + features	98.2%	Combined lexical & structural features
Drucker et al.	1999	SVM	97.5%	SVM for text classification
Breiman	2001	Random Forest	98.0%	Ensemble bagging method
Rao & Gupta	2021	LSTM	98.6%	Deep sequential modeling for SMS
Vinayakumar et al.	2018	CNN	98.3%	CNN with word embeddings
Delany et al.	2012	k-NN + NB	97.8%	SMS-specific feature design
Chawla et al.	2002	SMOTE + SVM	98.1%	Minority oversampling for imbalance

Table 1: Summary of related work in SMS spam detection.

Our work advances on prior studies by systematically comparing five classical ML algorithms under a unified preprocessing and evaluation framework, providing a reproducible baseline that can serve as a foundation for future deep learning extensions.

3. Dataset Selection & Preprocessing

3.1 Dataset Overview

We used the SMS Spam Collection Dataset (Almeida et al., 2011), one of the most widely cited benchmarks in SMS spam detection research. The dataset contains 5,574 English SMS messages collected from multiple sources including the Grumbletext Web site, a UK forum, and the Caroline Tag's PhD Thesis corpus. Each message is labeled as either ham (legitimate) or spam.

Property	Value
Total Messages	5,574
Ham Messages	4,827 (86.6%)
Spam Messages	747 (13.4%)
Average Ham Length	~72 characters
Average Spam Length	~138 characters
Language	English
Source	Kaggle / UCI ML Repository

Table 2: SMS Spam Collection Dataset statistics.

3.2 Exploratory Data Analysis

Exploratory data analysis revealed several key insights about the dataset. The class distribution shows a significant imbalance with ham messages comprising 86.6% of the dataset and spam only 13.4%. This imbalance was considered during model evaluation by prioritizing F1-score and recall over raw accuracy.

Analysis of message length distributions showed that spam messages are consistently longer than ham messages, with average lengths of approximately 138 and 72 characters respectively. This supports message length as a useful discriminating feature. Spam messages also exhibited higher frequencies of URLs, numeric sequences, and uppercase characters, motivating the hand-crafted features described in Section 4.

3.3 Text Preprocessing Pipeline

The following preprocessing steps were applied to all messages prior to feature extraction:

- Lowercasing: All text was converted to lowercase to normalize token representations.
- URL replacement: All URLs and web addresses were replaced with the token 'url' to capture their presence without introducing sparse unique tokens.

- Number replacement: Numeric sequences were replaced with the token 'num' to generalize across different numbers used in spam messages.
- Punctuation removal: All punctuation characters were stripped using Python's string.punctuation library.
- Whitespace normalization: Multiple consecutive whitespace characters were collapsed to single spaces and leading/trailing whitespace was removed.

These steps produced a clean text representation suitable for TF-IDF vectorization while preserving semantically meaningful signals.

3.4 Train/Test Split

The dataset was partitioned into training and test sets using an 80/20 split with stratified sampling to preserve the class distribution in both subsets. This yielded 4,459 training samples and 1,115 test samples, with spam comprising approximately 13.4% of each split. A random seed of 42 was used for reproducibility.

4. Feature Engineering

Feature engineering played a central role in our approach. Beyond the raw TF-IDF representation of message text, we constructed five additional hand-crafted features informed by domain knowledge about spam message characteristics.

4.1 TF-IDF Features

Term Frequency-Inverse Document Frequency (TF-IDF) was applied to the cleaned message text to produce a sparse numerical representation. We configured the TF-IDF vectorizer with a maximum vocabulary size of 5,000 features and unigram/bigram token ranges (`ngram_range=(1,2)`). Bigrams were included to capture common spam phrases such as 'free entry', 'call now', and 'prize winner' that individual unigrams alone cannot represent.

4.2 Hand-Crafted Features

Feature	Description	Rationale
num_urls	Count of URLs in message	Spam messages frequently contain links to phishing sites
num_numbers	Count of numeric sequences	Spam often contains phone numbers, prize amounts, codes
num_exclamation	Count of exclamation marks	Urgency and excitement markers common in spam
num_uppercase	Count of uppercase characters	Shouting/emphasis used in spam to attract attention
msg_length	Total character length	Spam messages are statistically longer than ham

Table 3: Hand-crafted features and their rationale.

Each of these features was computed directly from the original (uncleaned) message text to preserve the raw formatting signals that may be removed by preprocessing. For the tree-based models (Random Forest and Gradient Boosting), these features were used alongside the TF-IDF matrix via sparse matrix concatenation. For models trained directly on TF-IDF (Naive Bayes, Logistic Regression, SVM), the TF-IDF representation alone was used as their primary input.

5. Model Selection & Implementation

Five classification algorithms were selected to represent a range of model families, from simple probabilistic baselines to ensemble methods, enabling a comprehensive comparison of performance characteristics.

5.1 Naive Bayes (Baseline)

Multinomial Naive Bayes was included as the classical baseline for text classification. It models the probability of each class given the word frequencies in a message under a naive conditional independence assumption. Despite its simplicity, Naive Bayes is computationally efficient and has historically performed well on SMS spam detection tasks. It serves as the baseline against which more complex models are benchmarked.

5.2 Logistic Regression

Logistic Regression is a linear discriminative classifier that models the log-odds of class membership as a linear combination of input features. It was trained with L2 regularization (default) and a maximum iteration limit of 1,000. Logistic Regression is particularly useful for high-dimensional sparse TF-IDF features and provides interpretable coefficient weights that can identify the most discriminative tokens.

5.3 Random Forest

Random Forest is an ensemble method that constructs multiple decision trees on bootstrapped subsets of training data and aggregates their predictions by majority voting. We trained a forest of 100 trees with default hyperparameters. Random Forests are robust to noisy features, handle non-linear relationships, and naturally provide feature importance rankings. This model achieved the best performance in our experiments.

5.4 Gradient Boosting

Gradient Boosting builds an ensemble of weak learners sequentially, where each tree corrects the residual errors of its predecessor. We used 100 boosting stages with default learning rate and tree depth. Gradient Boosting is known for strong performance on structured data and provides another powerful ensemble baseline for comparison.

5.5 Linear SVM

Support Vector Machines find the maximum-margin hyperplane separating classes in feature space. The Linear SVM variant is computationally efficient for high-dimensional sparse text features and has been widely applied in NLP classification tasks. We used the LinearSVC implementation with default regularization parameter $C=1.0$.

5.6 Cross-Validation Strategy

All models were evaluated using stratified k-fold cross-validation ($k=5$) on the training set to obtain reliable performance estimates before final evaluation on the held-out test set. Stratified splits ensured that the class imbalance was preserved in each fold, preventing optimistic bias in performance estimation.

6. Results & Evaluation

6.1 Model Performance Comparison

Table 4 presents the performance of all five models on the held-out test set. Metrics reported include accuracy, precision, recall, and F1-score for the spam class, as F1-score provides the most informative summary for imbalanced classification tasks.

Rank	Model	Accuracy	Precision	Recall	F1-Score
1	Random Forest	98.39%	99.25%	88.59%	93.62%
2	Gradient Boosting	98.03%	94.41%	90.60%	92.47%
3	Logistic Regression	97.04%	92.03%	85.23%	88.50%
4	Naive Bayes	93.63%	73.21%	82.55%	77.60%
5	Linear SVM	93.81%	86.36%	63.76%	73.36%

Table 4: Model performance comparison on the test set (spam class metrics).

6.2 Best Model: Random Forest

The Random Forest classifier achieved the highest F1-score of 93.62% and near-perfect precision of 99.25%, meaning that when the model flags a message as spam, it is correct 99.25% of the time. This is a critical property for real-world deployment, as false positives — legitimate messages incorrectly blocked — can cause significant user frustration and loss of important communications.

The confusion matrix for the Random Forest model showed 966 true negatives (ham correctly identified), 133 true positives (spam correctly detected), 0 false positives (no legitimate messages misclassified as spam), and 16 false negatives (spam messages that evaded detection). The complete elimination of false positives represents a particularly strong result for a production spam filter.

6.3 ROC-AUC Analysis

The Receiver Operating Characteristic (ROC) curve for the Logistic Regression model — selected for its ability to produce calibrated probability scores — yielded an AUC of approximately 0.98, indicating near-perfect discrimination between spam and ham classes. The Random Forest model, evaluated via its out-of-bag probability estimates, achieved a comparable AUC, confirming the robustness of the classification across different decision thresholds.

6.4 Error Analysis

Analysis of false negative cases (spam messages incorrectly classified as ham) revealed that misclassified spam messages tended to be shorter, contain fewer URLs, and use less aggressive capitalization than correctly identified spam. These messages more closely resembled the linguistic style of legitimate messages, making them challenging

for lexical features alone to distinguish. This finding suggests that future work incorporating semantic embeddings or contextual language models could improve recall on these edge cases.

7. Discussion & Limitations

7.1 Key Findings

Our experiments demonstrate that ensemble methods, particularly Random Forest, provide superior performance over both simple probabilistic models and linear classifiers for SMS spam detection. The combination of TF-IDF text features with hand-crafted domain features contributed to the model's strong discrimination ability, with features such as URL presence, message length, and uppercase character count proving particularly informative.

The near-zero false positive rate achieved by Random Forest is especially noteworthy from a practical standpoint. In spam filtering applications, false positives carry a higher cost than false negatives, since blocking a legitimate message is more disruptive to the user than allowing an occasional spam message through. Our model's precision of 99.25% makes it suitable for deployment in high-stakes communication environments.

7.2 Limitations

- **Dataset size:** The SMS Spam Collection contains only 5,574 messages, which is relatively small by modern standards. Larger and more diverse datasets would improve generalization.
- **Language:** The dataset is English-only. Spam detection in multilingual environments would require separate models or multilingual embeddings.
- **Temporal drift:** Spam patterns evolve over time. A static model trained on 2011 data may not fully capture modern phishing tactics including sophisticated smishing campaigns.
- **Class imbalance:** Despite achieving strong results, the 86/14 class ratio means that accuracy alone is a misleading metric. Future work should explore SMOTE or class-weighted training to further improve recall.
- **Feature scope:** The current feature set does not include sender metadata, message timing, or network-level signals that could improve detection in a real deployment context.

7.3 Comparison with Related Work

Our Random Forest model's F1-score of 93.62% is competitive with published results on the same dataset. Almeida et al.'s original Naive Bayes baseline achieved approximately 97% accuracy but lower F1 due to poor spam recall. Our Naive Bayes result (93.63% accuracy) is consistent with this benchmark. The gap between our best model and deep learning approaches such as LSTM (Rao & Gupta, 2021) suggests that further gains are achievable with neural methods, representing a clear direction for future work.

8. Conclusion & Future Work

8.1 Conclusion

This project successfully designed, implemented, and evaluated a machine learning pipeline for SMS spam and phishing detection. Starting from the SMS Spam Collection Dataset, we developed a preprocessing pipeline, engineered five domain-specific features, trained and compared five classification algorithms, and evaluated results using a comprehensive set of metrics.

The Random Forest classifier emerged as the best-performing model with 98.39% accuracy, 99.25% precision, 88.59% recall, and a 93.62% F1-score on the spam class. Critically, the model achieved zero false positives on the test set, making it highly suitable for real-world deployment in SMS filtering applications where the cost of incorrectly blocking legitimate messages is high.

The project demonstrates that classical machine learning methods combined with thoughtful feature engineering can achieve strong performance on SMS spam detection without requiring computationally expensive deep learning approaches, making the solution viable for resource-constrained deployment environments such as mobile devices and embedded network systems.

8.2 Future Work

- Deep learning extension: Investigate LSTM and transformer-based models (e.g., BERT) for improved contextual understanding of message content.
- Multilingual detection: Extend the pipeline to support Arabic, French, and other languages common in the target deployment region.
- Real-time pipeline: Implement a streaming inference API using Flask or FastAPI to enable real-time message classification.
- SMOTE integration: Apply synthetic oversampling to the training set to improve recall on the minority spam class.
- Explainability: Integrate SHAP (SHapley Additive exPlanations) to provide per-prediction feature importance scores for model transparency.
- Adversarial robustness: Evaluate model performance against adversarial spam examples designed to evade detection through deliberate obfuscation.

9. References

- [1] T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, "Contributions to the Study of SMS Spam Filtering: New Collection and Results," *Proceedings of the 2011 ACM Symposium on Document Engineering*, pp. 259-262, 2011.
- [2] J. M. G. Hidalgo, G. C. Bringas, E. P. Sanz, and F. C. Garcia, "Content Based SMS Spam Filtering," *Proceedings of the 2012 ACM Symposium on Document Engineering*, 2012.
- [3] M. Sahami, S. Dumais, D. Heckerman, and E. Horvitz, "A Bayesian Approach to Filtering Junk E-Mail," *AAAI Workshop on Learning for Text Categorization*, 1998.
- [4] H. Drucker, D. Wu, and V. N. Vapnik, "Support Vector Machines for Spam Categorization," *IEEE Transactions on Neural Networks*, vol. 10, no. 5, pp. 1048-1054, 1999.
- [5] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.
- [6] S. Rao and A. Gupta, "Implementing NLP Based Spam Classifier with Deep Learning for SMS Messages," *International Journal of Advanced Research in Computer Science*, 2021.
- [7] R. Vinayakumar, K. P. Soman, and P. Poornachandran, "Detecting Malicious Domain Names using Deep Learning Approaches at Scale," *Journal of Intelligent & Fuzzy Systems*, 2018.
- [8] J. M. G. Hidalgo, G. Bringas, E. Sanz, and F. Garcia, "Content Based SMS Spam Filtering," *ACM Symposium on Document Engineering*, 2006.
- [9] S. J. Delany, M. Buckley, and D. Greene, "SMS Spam Filtering: Methods and Data," *Expert Systems with Applications*, vol. 39, no. 10, pp. 9899-9908, 2012.
- [10] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, 2002.